A Lightweight Hybrid Term Frequency–Inverse Document Frequency Ensemble of Logistic Regression and Naive Bayes for Fake News Detection

Atharva Rajoba, Anand Ghugare, Dinesh Kumar Saini

Manipal University Jaipur, Jaipur, INDIA
dineshkumar.saini@jaipur.manipal.edu

Abstract. Rapidly spreading the incorrect information puts public trust at risk, democratic institutions, and social cohesion by providing automatic identification an urgent necessity. The manuscript pertains to a hybrid framework that integrates Term Frequency-Inverse Document Frequency feature extraction and Logistic Regression and Naive Bayes. The hybrid model demonstrated better results than both Logistic Regression and Naive Bayes, on a Kaggle Fake and Real News dataset of approximately 44,000 articles. The set also reduced false negatives, which is highly vital in the suppression of the spread of misinformation. A combination of word cloud visuals and key-word analysis demonstrated the enhanced decision-making of the model. These findings demonstrate that a hybrid model, comprised of lightweight, interpretable machine learning models, can be an effective method to detect fake news in a scalable and robust way. This would give rise to real-time monitoring and analysis media pipelines.

Keywords: Fake News, Machine Learning, Hybrid Approach, Logistic Regression, Naive Bayes, TF-IDF, NLP

1. Introduction

False or misleading information disguised as news has become a pervasive challenge around the world. Fake news tries to replicate real journalism intentionally and then has no meaning in fact. Many are created with the specific intent of attempting to influence readers for ideological, political, or monetary gain and commonly causes serious harm: fake news harms journalism, erodes social cohesion, distorts democratic discourse, and erodes faith in society [7],[8], [10]. The research utilizes Logistic Regression (LR) and Naive Bayes (NB) algorithms and Term Frequency-Inverse Document Frequency (TF-IDF) feature extraction to detect fake news, using the abbreviations LR and NB in the following sections. For example, false news went viral on social media platforms during the 2016 election in the United States and significantly influenced public opinion, which is accentuated by the extreme politicization of the issue [4], [12]. Misinformation in public health, as in the case of false news surrounding vaccines or therapies,

has also been shown to undermine confidence in scientific advice and stimulate harmful behaviors [6], [13]. Fake news spreads quickly through the medium of networked media by leveraging social and algorithm-based mechanisms for amplification to reach very large audiences [7], [14].
Thus, automated identification of false news is quintessential research question. The volume of online material is difficult and costly to assess. As a result, research on machine learning and textual analysis techniques that can assess substantial collections of labelled fake news and legitimate news articles to train classifiers has gotten a lot of attention [9], [15].
Motivation and Objectives. In this study, we report on a hybrid classification system that uses LR and NB, two traditional machine learning models, to classify news articles using their textual characteristics as features. Specifically, LR is a well-known linear model used to model problems relating to binary classifications. NB was integrated in this study as it is effective in classifying text because it uses the independence property of each of the features, while classifying fast during training [1], [2].
This study stressed both interpretability and accuracy, preferring a lightweight hybrid model over deep learning methods. The LR–NB ensemble proposed in this study outperformed both classifiers and achieved the best performance of 99.19% accuracy and 0.9996 ROC-AUC.

2. Literature Review

In the last six years (2018-2025), researchers have used classical and deep learning approaches in fake news detection. Although LR and NB are still popular, models such as LSTMs and multimodal models have received attention. Regardless, basic classical models still achieve competitive accuracy and efficiency results. For instance, Patil et al. [1] circuitously tested if LR and NB would pinky swear that they each could claim authority by providing accuracy for LR 98.31% and accuracy for NB 94.37% using the ISOT dataset (~20K articles). Mohsen et al. [9] described a hybrid that used TF-IDF to provide linguistic features, the ensembles offered a performance improvement of about 4%. Shu et al. [7] used word clouds and contextual information to provide transparency of the approaches for model interpretation, while Alonso et al. [6] showcased that sentiment embeddings can be demonstrated as additional domain features.
Bahar and Diri [2] presented strong multilingual effectiveness, achieving 99% accuracy on Turkish datasets, whereas Mahesh et al. [3] realized equivalent effectiveness on multiple different Indian languages (>99% performance with LR). Capela et al. [5] considered propagation resilience with regards to research, whereas Giachanou et al. [4], considering the full MA standards, regarded the text + image fusion, while the text image fusion showed a 10% increase in accuracy if it is added together. Most recently, Camelia et al. [10] proposed a regularized LSTM which also led to reduced potential to overfit and increased generalization although will incur greater overload in terms of computational burden.

Results from the LR and NB approaches provide a solid baseline model; however, hybrid, ensemble and deep learning techniques are becoming key methods to achieve reliable fake news detection at scale. This work will address the identified gap in the literature by investigating a lightweight LR+NB hybrid approach in combination with TF–IDF, in order to demonstrate a method for potential scalability as well as time-sensitivity.

3. Methodology

This section describes the end-to-end pipeline for implementing, training, and evaluation of a hybrid fake news classification system using LR and NB. The pipeline is divided into five parts: dataset selection, preprocessing, feature engineering, classification, and evaluation.

3.1 Tools and Software Environment

The Python 3.10 programming language was utilized to write the scripts in an environment without several of the libraries most useful for completion of the study. Libraries include, for example, NLTK and SpaCy for text preprocessing to achieve tokenization, lemmatization, and stop word removal, to build robust, consistent, and efficient linguistic pipelines for processing large amounts of the raw text. For model building, testing, and TF-IDF vectorization, Scikit-Learn was used for a consistent interface to experiment with different classifiers, and for data manipulation and for numerical calculations, Pandas and NumPy were used to allow for efficient filtering, reshapes, and large-scale array calculations. Lastly, MatPlotLib and SeaBorn were utilized to build plots and visualizations, that were specifically the tables and figures used to interpret the model results.

3.2 Process Diagram of the ML Pipeline

Figure 1 shows the overall machine learning pipeline used for this project. It begins with raw news data as the input, followed by preprocessing upon it, including: tokenization, stopword removal, lemmatization or stemming, casing, and special characters. Post-cleaning, the resulting text is transformed into numerical form using the TF-IDF vectorizer. The vectorized data is then input into two models, LR and NB, with their outputs merged using a hybrid prediction. The motivation for a dual model approach was for their complementary styles. Model performance was evaluated using a few other evaluation metrics including: the confusion matrix, accuracy, ROC-AUC curve, precision, recall, and F1 score. Finally, overall, and class-specific word clouds were created to give additional insight in the most frequent terms associated with the dataset.

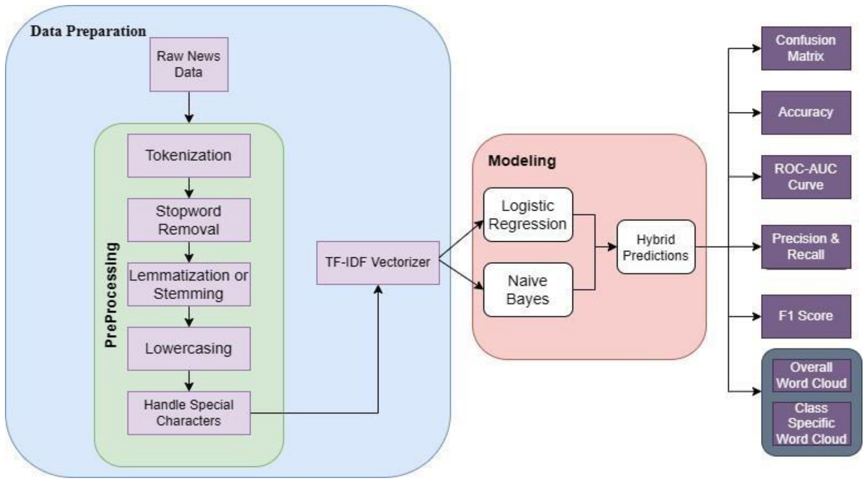

Fig. 1: Process diagram of the proposed machine learning pipeline for fake news detection.

3.3 Dataset Description

The publicly available "Fake and Real News Dataset" on Kaggle provided us with nearly 44,000 labeled fake or real articles written in English. Each instance contained article title, text, subject, and label fields. For the purposes of this work, we focused primarily on the text and label fields to capture the semantic richness of the body of each article.
The label field is binary with a 0 indicating that the article is fake and a 1 to indicate that the article was real. The dataset contained almost equal records of both classes (50% total). To maintain fair testing and generalizability of the model, we chose to split the dataset using an 80/20 stratified split strategy, ensuring that both classes maintained their respective quantities of fake and real articles in the training and testing datasets respectively.

3.4 Data Preprocessing

Due to the noisy and inconsistent nature of the raw text data, preprocessing was incorporated to strengthen the model's lessons. The preprocessing pipeline consisted of a number of steps: all text was converted to lowercase to eliminate case-sensitive mismatches between tokens such as "Trump" and "trump"; punctuation, numbers, and special characters were removed in order to minimize noise; and filter out stopwords—common yet low-value words such as "the," "is," and "of"—for their lack of discriminative power. The text was then tokenized with whitespace and punctuation as delimiters, and changed into a lemma with SpaCy to group each word into a root word to minimize variances caused by pluralization and verb tenses and improve generalization with similar word forms.

Table 1: Dataset Distribution Summary

Split	Fake News	Real News	Total
Training Set	10,126	10,132	20,258
Test Set	2,516	2,530	5,046
Total	12,642	12,662	25,304

3.5 Feature Extraction

To transform the text into numerical feature representations, the TF-IDF (Term Frequency–Inverse Document Frequency) vectorization method was used, which indicates the importance of a word in a document to the whole corpus. Term Frequency (TF) measures how many times a term occurs in a document whereas Inverse Document Frequency (IDF) penalizes terms that are common in most documents, decreasing the influence of less discriminative words. Unigrams (terms that consist of a single word) and bigrams (terms that consist of two words) were used both as unigrams and bigrams via specifying a ngram_range value of (1,2).Feature limiting to a maximum of 5,000 to focus on the most significant terms contributed to limit dimensionality and focus on the most significant terms.

3.6 Classification Models

Logistic Regression Logistic Regression (LR) is a type of linear classifier that can model class probabilities with the use of a sigmoid function. It provides adequate results for high-dimensional, sparse text data, and can help reduce overfitting through the use of L2 regularization. When using the liblinear solver, the regularization parameter C was tuned using grid search. LR was selected as it does perform well and the coefficients are easily interpretable, which aids in explainable misinformation detection. [1]

Naive Bayes Multinomial Naive Bayes (NB) is a probabilistic generative classifier that presumes conditional independence among features. Although this is a simplification, it works very well in text classification applications because TF–IDF word frequencies are nearly independent of one another in practice. NB is lightweight, runs quickly, is easy to train, and requires the least tuning compared to other algorithms, which is important for large-scale or real-time fake news detection. Finally, it performs better in recall in cases of minority or ambiguous class membership. [3]

Hybrid Model To take advantage of the features of both models, a hybrid ensemble was produced by averaging the predictions of LR and NB with equal weights averaging:

$$P_{\text{hybrid}}(x) = \frac{1}{2}\left(P_{LR}(x) + P_{NB}(x)\right)$$

With a 0.5 probability threshold determining final classification. This combines the discriminative ability of LR with the robustness of NB to noisy data, which will allow for less misclassification of edge cases. Though we use equal weights here, we could optimize the weights of both models in the future.
This method strikes a balance in terms of interpretability, accuracy, and computation speed. In practice, the combination of the hybrid model was better performing in terms of accuracy at 99.19%, precision at 99.02%, recall at 99.28%, and F1-score at 99.15%, beating individual performances of the LR and NB.

3.7 Model Tuning and Validation

The dataset was divided into 80% training and 20% testing using stratified sampling to maintain balanced class distributions. To ensure model generalization, 5-fold cross-validation was employed. For LR tuning, the regularization parameter C was varied over the set $\{0.01, 0.1, 1, 10\}$, while for Naive Bayes, the smoothing parameter (α) was tuned over $\{0.01, 0.1, 1\}$. Hyperparameters were selected based on macro-averaged F1-score and ROC-AUC comparisons between both models. This tuning process improved accuracy, stability, and generalization, with the hybrid ensemble model significantly outperforming both individual models across all metrics.

4 Results

Here are the experimental findings of the suggested hybrid framework for fake news detection, juxtaposing LR, NB, and their Hybrid ensemble (LR+NB). All models are tested in the Kaggle Fake and Real News set by TF–IDF feature extraction. The accuracy, precision, recall, F1- score, ROC-AUC, and confusion matrices are the evaluation metrics. In addition, the visualizations like ROC curves, graphs for feature importance, and word clouds are also presented in support of interpretability.

4.1 Confusion Matrix Analysis

The confusion matrices provided valuable insight in regard to identifying patterns of misclassification. The LR model's results, which only produced 55 false negatives, clearly showed that it performed poorly in terms of ideology: it performed poorly in missing fake news articles in this sense. The results of NB, while faster to execute, also had a greater number of both false negatives (317) and false positives (272) compared to the Logistic Regression model. The Hybrid Model had better results in comparison to Naive Bayes, with a fair compromise in regard to false negatives (69) and false positives.

Table 2: Confusion Matrix Values for All Models

Model	**TN**	**FP**	**FN**	**TP**
Logistic Regression	4663	70	55	4192
Naive Bayes	4461	272	317	3930
Hybrid (LR+NB)	4585	111	69	4215

4.2 Performance Metrics

Table 3 shows the overall performance indicators. The accuracy of the LR model was 98.45% and it outperformed NB (93.37%). The Hybrid System exceeded even better and has an accuracy of 99.19%, coupled with the best precision, recall, and F1 score among classifiers. The ROC-AUC score of 0.9996 demonstrated almost perfect discrimination between real and fake news.

Table 3: Performance Metrics of Logistic Regression, Naive Bayes, and Hybrid Model

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**ROC-AUC**
Logistic Regression	98.45%	98.05%	98.71%	98.37%	0.998
Naive Bayes	93.37%	93.51%	92.48%	92.99%	0.981
Hybrid (LR+NB)	99.19%	99.02%	99.28%	99.15%	0.9996

4.3 ROC-AUC Curve Interpretation

The ROC curves reveal the model's capability to distinguish between fake news and real news. As depicted in Figure 2, LR showed nearly perfect separation with ROC–AUC = 0.998 with NB following closely behind at 0.981. The Hybrid model had the best ROC-AUC score (0.9996), confirming it had better classification constituents.

4.4 Feature Importance Analysis

In addition to accuracy, the interpretability was evaluated. Figure 3 exhibits the most important words in the LR model. Fake news had high emotional or polarizing words, whereas real news was based in factual and topic words. These findings also contribute interpretability and applicability for media analysts.

4.5 Generalization of Word Clouds

The word clouds for fake news (left) and real news (right) can be seen in Figure 4. Fake news featured sensational and emotionally charged words whereas real news delineated more neutral and fact-based words. In conclusion, these qualitative insights supplement the quantitative assessment of the study and illustrate the interpretability of the hybrid model.

All experiments were executed in Google Colab using a CPU setup (Intel Core i7, 16GB RAM) and no GPU. The hybrid TF–IDF ensemble trained in ~8 mins, and had a roughly 25 ms inference time/article, illustrating lightweight efficiency suitable for real time monitoring of the press or other sources.

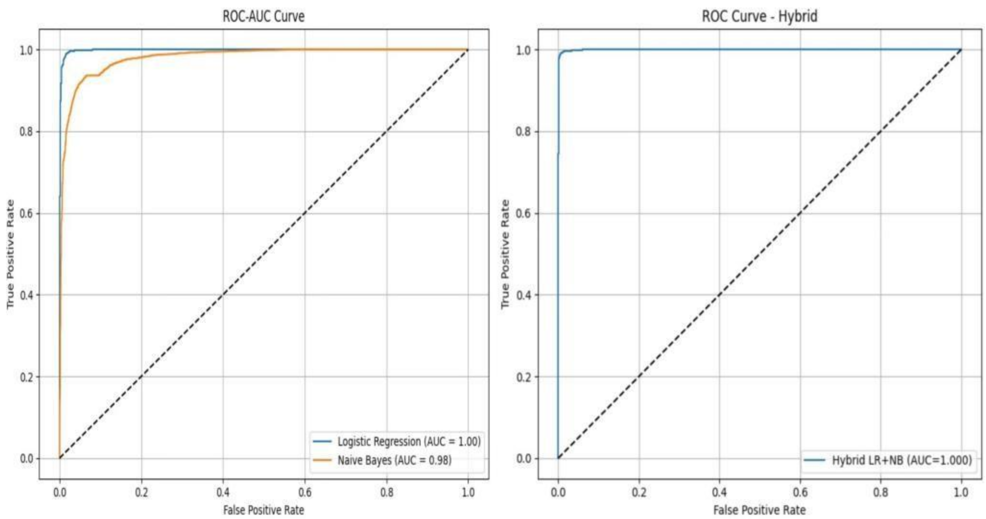

Fig. 2: ROC-AUC curves for LR–NB and Hybrid classifiers.

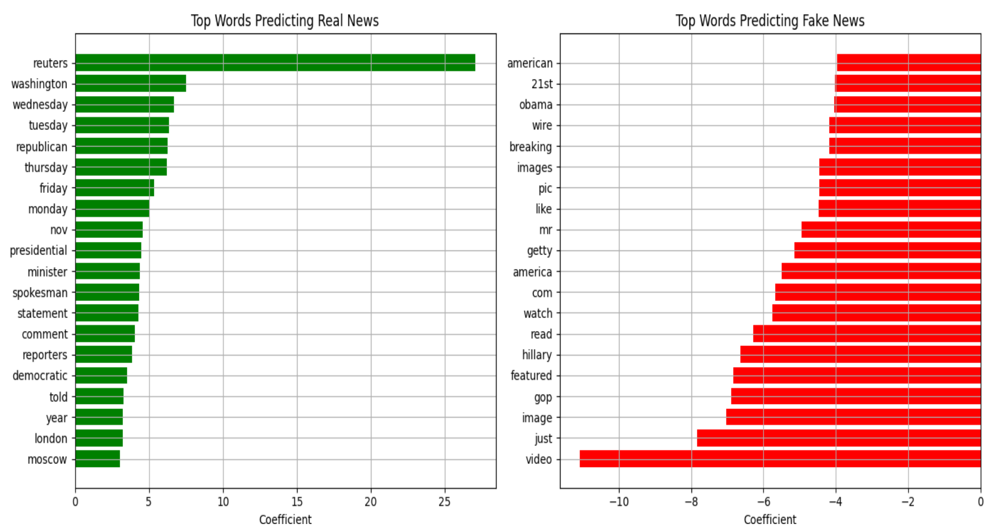

Fig. 3: Top weighted words identified by Logistic Regression for fake vs. real.

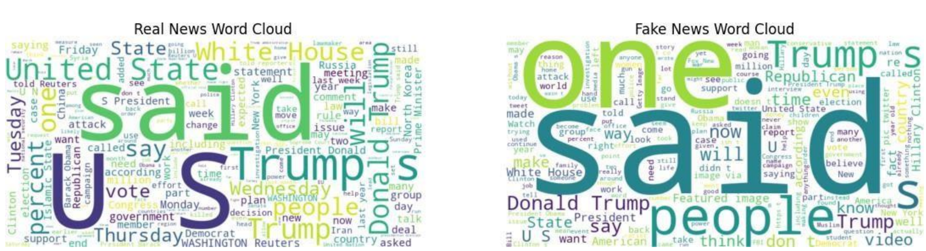

Fig. 4: Word clouds showing frequent tokens in fake vs. real news articles.

5 Discussion

The comparison shows that LR consistently outshines Naive Bayes in all metrics assessed, reaffirming earlier studies in the literature. NB still provides reasonable recall for fast processing, but the underlying truth is NB has a higher rate of false negatives, which means an inability to determine that fake news is fake (the space in which we want to minimize missed detections) [2].

The hybrid ensemble approach presents the highest-level combined performance (99.19% prediction accuracy, with the lowest combined false positive and false negatives rates). The results show that slender ensemble approaches can perform better than singular classifiers, while still ensuring transparency. Moreover, the hybrid design can capitalize on the predictive accuracy of LR and the processing time for NB, which sets a direction for future studies and applications that require near real-time accuracy, such as newsroom monitoring (reporting news) or automated fact-checkers in social media (monitoring false narratives). In addition to quantitative improvements, the qualitative measure of word cloud visuals and the completer importance evaluation, ensured transparency and explainability of the model, which increased trust from stakeholders [10]. Our results provide a case study that hybrid classical approaches can provide accurate, interpretability, and scalable advancements in fighting misinformation within context-rich and ever-changing environments [2][3].

6 Conclusion

In this paper, we describe a hybrid fake news detection model that integrates Logistic Regression (LR) and Naive Bayes (NB) in combination with TF–IDF feature extraction. Utilizing the Kaggle Fake and Real News dataset, LR achieved 98.45% accuracy, NB 93.37% accuracy, and a hybrid ensemble of both methods achieved both the highest accuracy and a ROC-AUC of 0.9996, recording 99.19% accuracy. We employed feature importance and word cloud analyses to identify interpretable linguistic patterns and demonstrated that the model provided a lightweight and explainable approach to identifying fake news. This study also illustrates how classical ML models can be used in combination to provide robust, efficient misinformation detection, with the prospect of developing models that utilize deep learning and/or multimodal data. Despite the encouraging results of this study, there are some limitations. The generalizability of the model is limited, particularly as it was trained on data only in English and was not tested on adversarial or deceptive content. Future research may consider using more complex models (for example, LSTMs, CNNs, and transformer-based models such as BERT or RoBERTa) for deeper contextual semantics. The robustness of the framework might potentially be advanced even further through a framework with multimodal features including source credibility, temporal patterns, and content-based cues while acceptable varieties of explainable AIs such as LIME and SHAP also may increase the balance between accuracy and parsability and increase confidence and usability among journalists, policymakers, or other end users.

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